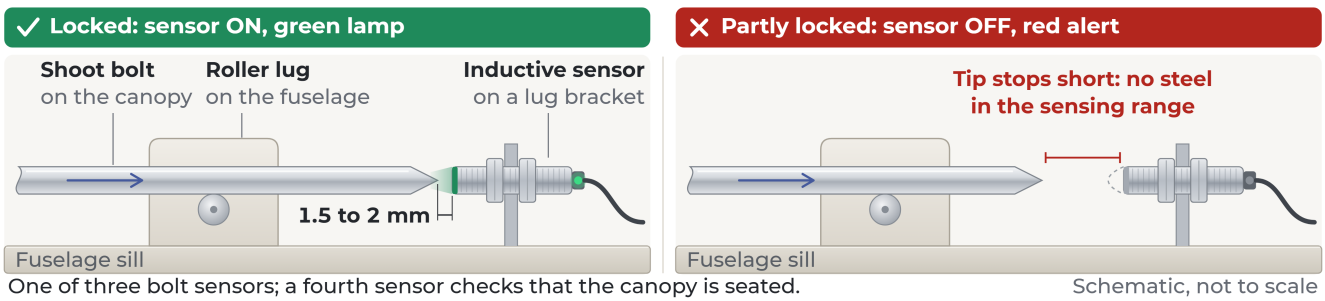
Table 1. For each element we kept the option that reaches a busy pilot and never takes away engine power.

Element	Choice	Why not the others
Sensing point	Each bolt and the seat	One pin (the task-sheet example), the handle or the linkage cannot see one bolt out (Table 4)
Sensor type	Line-monitored inductive sensor	Snap switches, Hall sensors and bare contacts give more false alerts (Table 4); a worn or wet bare contact, our first idea, can read ”locked”
Aural	Voice without a lead tone	A lead tone slowed responses [12]
Vibration	Lap-belt pad	A stick shaker means ”stall”; a knob shaker vibrates an engine control, which must not creep under vibration [13]. The waist matched the hand’s response time and, with a strong vibration, half the wrong actions (10.9 % against 22.3 %) [14]
Barrier	None; alerts only	Removed for safety: engine shut-down, a take-off gate and a throttle pawl act on the engine or throttle, so a fault could take away power in flight
G3X link	Unit output into the G3X	A raw switch on the two-state G3X input [15] would read a short to ground as ”closed”

Text.



One of three bolt sensors; a fourth sensor checks that the canopy is seated. Schematic, not to scale

Figure 1. Each sensor sees the steel bolt tip only at full lock; a partly locked bolt leaves it off, and the unit shows red.